

BHARATHWAJ RAMANATHAN

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EDUCATION

University of Illinois Urbana-Champaign

MEng, Mechanical Engineering | GPA: 3.91/4.00

Aug 2024 – Dec 2025

Champaign, IL

PSG College of Technology

BE, Mechanical Engineering | GPA: 8.59/10.00

Jul 2020 – May 2024

Coimbatore, India

EXPERIENCE

Mechanical Engineering Intern | Composite Structures

Mar 2026 – Present

Vinjamuri Innovations

New Rochelle, NY

- Built a nonlinear **hybrid solid-shell FEA** model of a **Type III COPV** in **ANSYS 2025 R2**, predicting burst pressure within **14%** of confirmed **1080 bar** by simulating **3-step autofrettage** at **593 bar** for a **ruggedized high-pressure** hydrogen storage system.
- Confirmed design safety against **ISO 11119-2** requirements up to **400 bar** service by validating residual compressive stress from autofrettage, supporting structural integrity documentation for a defense-adjacent high-pressure application.
- Validated **7-ply CFRP/GFRP** hoop stress distribution in **ANSYS ACP** against published literature by modeling composite layup from CNC winding data, confirming load sharing behavior post-autofrettage for a **ruggedized pressure vessel** application.

Graduate Research | Thermal & Mechanical Design

Jan 2025 – Dec 2025

University of Illinois Urbana-Champaign

Champaign, IL

- Designed and evaluated **BCC lattice cold plate** architectures for **electronics thermal management**, running **30 CFD simulations** to map cooling efficiency vs. hydraulic resistance across geometry variants and produce a validated thermal design space.
- Reduced design iteration time by **100+ hours** by building fast-running **MATLAB** models to predict T_{max} and pressure drop across the design space, enabling rapid identification of the optimal **cold plate** configuration without additional CFD runs.

Mechanical Design Engineer Intern

Jun 2025 – Aug 2025

Vinjamuri Innovations

New Rochelle, NY

- Reconstructed full assembly CAD of a patented **Type III composite hydrogen vessel** in **SolidWorks** from drawings only, delivering a simulation-ready model that enabled structural analysis of a **safety-critical ruggedized pressure vessel** ahead of physical testing.
- Identified safe operating margins across **160–750 bar** by performing **nonlinear FEA** on stress and deformation response, providing the structural verification basis for design sign-off on a high-pressure hydrogen storage system.

Mechanical Engineering Intern | Capstone Project

Dec 2023 – May 2024

Prenac Tools

Coimbatore, India

- Reduced material replacement costs by **30%** by redesigning a collapsible industrial pallet in **SolidWorks**, validated to a **1-ton** load capacity via **FEA** through geometry improvements targeting structural durability and reuse cycles.
- Improved prototype design through iterative sim-vs-test comparison by leading full fabrication including sheet metal cutting, welding, and **GD&T**-controlled assembly per **ASME Y14.5**, then using physical load test discrepancies to drive design corrections.

Project Trainee | Structural Analysis

May 2023 – Jun 2023

Defence Research & Development Organisation (DRDO) | CABS

Bengaluru, India

- Verified structural integrity of an **avionics electronics rack** under **shock and vibration** loading for the **Embraer ERJ-145**, maintaining deformation below **3.13 mm** and stress below **238 MPa** at **2.56 g** through limit and emergency conditions up to **9 g** on a defense airborne platform.

PROJECTS

EasyBlot: Automated Western Blotting Instrument

Aug 2025 – Dec 2025

SolidWorks · Custom 2-Layer PCB · ESP32 · Peristaltic Fluidics

- Designed and built a full **electromechanical system** from scratch, integrating a machined enclosure, fluidic routing, custom **2-layer PCB**, and **ESP32** embedded control across mechanical, electrical, and firmware subsystems into a single **functional prototype**.
- Automated timed reagent sequencing across a **6-pump peristaltic fluidic system** via firmware, cutting active user lab time by **4–6 hours** per run and eliminating operator variability from manual protocols.

Immersion-Cooled EV Battery Module

Aug 2024 – Dec 2024

SolidWorks · ANSYS Icepak

- Achieved **46%** reduction in peak cell temperature and cut thermal variation from **23.7 °C** to **7.0 °C** by redesigning flow geometry and cell arrangement for a 9-cell cylindrical module, validated in **ANSYS Icepak** as a **thermal management system** for high-density energy storage.
- Selected **EG 50-50** as the optimal coolant from 6 dielectric and coolant candidates evaluated on thermal conductivity and viscosity tradeoffs, reducing system pressure drop while improving thermal uniformity and demonstrating **TMS design** judgment beyond simulation alone.

TECHNICAL SKILLS

CAD & Design: SolidWorks, Fusion 360, nTopology, GD&T (ASME Y14.5), Tolerance Analysis, DFMA

Simulation: ANSYS Mechanical, Abaqus, Hypermesh, ANSYS Icepak, ANSYS ACP

Manufacturing: Sheet Metal Design, CNC Machining, Composite Layup, Rapid Prototyping (3D Printing)

Programming: Python, MATLAB, Data-Driven Design, KiCad (Custom PCB), ESP32